THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **Areas Related to Circles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 56 cm²	Circle $= 154$ cm^2; square $= 98$ cm^2; Difference $= 56$ cm^2
2	**(d)** 672 cm²	Circle area $= \frac{22}{7} \times 28^2 = 2464$ cm^2; square $= 3136$ cm^2; Left over $= 672$ cm^2
3	**(a)** 1034 m²	Area $= \pi(R^2 - r^2) = \frac{22}{7}(27^2 - 20^2)$; Area of path $= 1034$ m^2
4	**(d)** 77 cm²	$\frac{60}{360} \times \frac{22}{7} \times (14^2 - 7^2)$; Area $= 77$ cm^2
5	**(c)** 90°	$\theta = \frac{22 \times 360}{2 \times \frac{22}{7} \times 14}$; $\theta = 90^\circ$
6	**(d)** 686 cm²	Sector $= 1886.5$ cm^2, triangle $= \frac{1}{2} \times 49 \times 49 = 1200.5$ cm^2; Segment $= 1886.5 - 1200.5 = 686$ cm^2
7	**(a)** 1089 cm²	Side $= 132 \div 4 = 33$ cm; Area $= 33^2 = 1089$ cm^2
8	**(b)** 42 cm²	Quadrant $= 154$ cm^2; square $= 196$ cm^2; Remaining $= 42$ cm^2
9	**(d)** 224 cm²	Circle $= 616$ cm^2; square $= 392$ cm^2; Difference $= 224$ cm^2
10	**(d)** 1050 cm²	Circle area $= \frac{22}{7} \times 35^2 = 3850$ cm^2; square $= 4900$ cm^2; Left over $= 1050$ cm^2

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	In 15 minutes the hand turns through $6^\circ \times 15 = 90^\circ$; Area $= \frac{90}{360} \times \frac{22}{7} \times 28^2$ (1); Area $= 616$ cm^2 (1).	1+1
12	The horse can graze a sector of angle 180° and radius 14 m; Area $= \frac{180}{360} \times \frac{22}{7} \times 14^2$ (1); Area $= 308$ m^2 (1).	1+1
13	Circumference $= 2 \times \frac{22}{7} \times 35 = 220$ cm; Distance 55 m $= 5500$ cm; revolutions $= \frac{5500}{220}$ (1); Revolutions $= 25$ (1).	1+1
14	In 30 minutes the hand turns through $6^\circ \times 30 = 180^\circ$; Area $= \frac{180}{360} \times \frac{22}{7} \times 28^2$ (1); Area $= 1232$ cm^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Circumference $= 2\pi r = 2 \times \frac{22}{7} \times 14 = 88$ m (1); Cost $= 88 \times 12$ (1); Cost $=$ ₹ 1056 (1).	1+1+1
16	Let the new radius be R; $2\pi R = 2\pi r_1 + 2\pi r_2$ (1); $R = r_1 + r_2 = 14 + 7$ (1); $R = 21$ cm (1).	1+1+1
17	Let the new radius be R; $\pi R^2 = \pi r_1^2 + \pi r_2^2$ (1); $R^2 = 3^2 + 4^2 = 25$ (1); $R = 5$ cm (1).	1+1+1
	OR: Circumference $= 2\pi r = 2 \times \frac{22}{7} \times 21 = 132$ m (1); Cost $= 132 \times 9$ (1); Cost $=$ ₹ 1188 (1).	1+1+1
18	Let the new radius be R; $2\pi R = 2\pi r_1 + 2\pi r_2$ (1); $R = r_1 + r_2 = 18 + 11$ (1); $R = 29$ cm (1).	1+1+1
	OR: Let the new radius be R; $\pi R^2 = \pi r_1^2 + \pi r_2^2$ (1); $R^2 = 8^2 + 15^2 = 289$ (1); $R = 17$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Required area $= \frac{\theta}{360} \times \pi(R^2 - r^2)$ (2); $\frac{45}{360} \times \frac{22}{7} \times (14^2 - 7^2)$ (2); Area $= 57.75$ cm^2 (1).	2+2+1
	OR: The 8 ribs divide the circle into 8 equal sectors (2); Area of the circle $= \frac{22}{7} \times 42^2 = 5544$ cm^2 (2); Area between ribs $= 5544 \div 8 = 693$ cm^2 (1).	2+2+1
20	Required area $= \frac{\theta}{360} \times \pi(R^2 - r^2)$ (2); $\frac{72}{360} \times \frac{22}{7} \times (14^2 - 7^2)$ (2); Area $= 92.4$ cm^2 (1).	2+2+1
	OR: The 14 ribs divide the circle into 14 equal sectors (2); Area of the circle $= \frac{22}{7} \times 35^2 = 3850$ cm^2 (2); Area between ribs $= 3850 \div 14 = 275$ cm^2 (1).	2+2+1

Mistakes to avoid:

- Using πr^2 in the arc-length formula or $2\pi r$ in the sector-area formula.
- Forgetting to add the two radii when asked for the *perimeter* of a sector.
- Using diameter in place of radius in πr^2.
- Subtracting the wrong figure for a segment: minor segment $=$ sector $-$ triangle, not sector $+$ triangle.
- Mixing units (cm and m) or writing cm instead of cm^2 for areas.
- Thinking the area of a circle doubles when the radius doubles; it becomes four times.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **Areas Related to Circles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 814 m²	Area $= \pi(R^2 - r^2) = \frac{22}{7}(22^2 - 15^2)$; Area of path $= 814$ m²
2	**(b)** 154 cm²	$\frac{120}{360} \times \frac{22}{7} \times (14^2 - 7^2)$; Area $= 154$ cm²
3	**(a)** 45°	$\theta = \frac{11 \times 360}{2 \times \frac{22}{7} \times 14}$; $\theta = 45^\circ$
4	**(d)** 896 cm²	Sector $= 2464$ cm², triangle $= \frac{1}{2} \times 56 \times 56 = 1568$ cm²; Segment $= 2464 - 1568 = 896$ cm²
5	**(b)** 3025 cm²	Side $= 220 \div 4 = 55$ cm; Area $= 55^2 = 3025$ cm²
6	**(d)** 168 cm²	Quadrant $= 616$ cm²; square $= 784$ cm²; Remaining $= 168$ cm²
7	**(c)** 504 cm²	Circle $= 1386$ cm²; square $= 882$ cm²; Difference $= 504$ cm²
8	**(b)** 514.5 cm²	Circle area $= \frac{22}{7} \times 24.5^2 = 1886.5$ cm²; square $= 2401$ cm²; Left over $= 514.5$ cm²
9	**(c)** 1474 m²	Area $= \pi(R^2 - r^2) = \frac{22}{7}(37^2 - 30^2)$; Area of path $= 1474$ m²
10	**(a)** 115.5 cm²	$\frac{90}{360} \times \frac{22}{7} \times (14^2 - 7^2)$; Area $= 115.5$ cm²

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The horse can graze a sector of angle 180° and radius 21 m; Area $= \frac{180}{360} \times \frac{22}{7} \times 21^2$ (1); Area $= 693$ m² (1).	1+1
12	Circumference $= 2 \times \frac{22}{7} \times 14 = 88$ cm; Distance 88 m $= 8800$ cm; revolutions $= \frac{8800}{88}$ (1); Revolutions $= 100$ (1).	1+1
13	In 45 minutes the hand turns through $6^\circ \times 45 = 270^\circ$; Area $= \frac{270}{360} \times \frac{22}{7} \times 28^2$ (1); Area $= 1848$ cm² (1).	1+1
14	The horse can graze a sector of angle 180° and radius 28 m; Area $= \frac{180}{360} \times \frac{22}{7} \times 28^2$ (1); Area $= 1232$ m² (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Circumference $= 2\pi r = 2 \times \frac{22}{7} \times 35 = 220$ m (1); Cost $= 220 \times 6$ (1); Cost $=$ ₹ 1320 (1).	1+1+1
16	Let the new radius be R; $2\pi R = 2\pi r_1 + 2\pi r_2$ (1); $R = r_1 + r_2 = 25 + 13$ (1); $R = 38$ cm (1).	1+1+1
17	Let the new radius be R; $\pi R^2 = \pi r_1^2 + \pi r_2^2$ (1); $R^2 = 12^2 + 16^2 = 400$ (1); $R = 20$ cm (1).	1+1+1
	OR: Circumference $= 2\pi r = 2 \times \frac{22}{7} \times 28 = 176$ m (1); Cost $= 176 \times 11$ (1); Cost $=$ ₹ 1936 (1).	1+1+1
18	Let the new radius be R; $2\pi R = 2\pi r_1 + 2\pi r_2$ (1); $R = r_1 + r_2 = 30 + 12$ (1); $R = 42$ cm (1).	1+1+1
	OR: Let the new radius be R; $\pi R^2 = \pi r_1^2 + \pi r_2^2$ (1); $R^2 = 15^2 + 20^2 = 625$ (1); $R = 25$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Required area $= \frac{\theta}{360} \times \pi(R^2 - r^2)$ (2); $\frac{150}{360} \times \frac{22}{7} \times (14^2 - 7^2)$ (2); Area $= 192.5$ cm² (1).	2+2+1
	OR: The 9 ribs divide the circle into 9 equal sectors (2); Area of the circle $= \frac{22}{7} \times 21^2 = 1386$ cm² (2); Area between ribs $= 1386 \div 9 = 154$ cm² (1).	2+2+1
20	Required area $= \frac{\theta}{360} \times \pi(R^2 - r^2)$ (2); $\frac{108}{360} \times \frac{22}{7} \times (14^2 - 7^2)$ (2); Area $= 138.6$ cm² (1).	2+2+1
	OR: Grazed area is a quadrant: $\frac{1}{4}\pi r^2$ (2); Increase $= \frac{1}{4} \times \frac{22}{7}(28^2 - 7^2)$ (2); Increase $= 577.5$ m² (1).	2+2+1

Mistakes to avoid:

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- Forgetting to add the two radii when asked for the *perimeter* of a sector.
- Using diameter in place of radius in πr^2.
- Subtracting the wrong figure for a segment: minor segment $=$ sector $-$ triangle, not sector $+$ triangle.
- Mixing units (cm and m) or writing cm instead of cm² for areas.
- Thinking the area of a circle doubles when the radius doubles; it becomes four times.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **Areas Related to Circles** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 180°	$\theta = \frac{44\times360}{2\times\frac{22}{7}\times14}$; $\theta = 180^\circ$
2	**(c)** 1134 cm²	Sector $= 3118.5$ cm^2, triangle $= \frac{1}{2}\times63\times63 = 1984.5$ cm^2; Segment $= 3118.5 - 1984.5 = 1134$ cm^2
3	**(d)** 5929 cm²	Side $= 308 \div 4 = 77$ cm; Area $= 77^2 = 5929$ cm^2
4	**(c)** 262.5 cm²	Quadrant $= 962.5$ cm^2; square $= 1225$ cm^2; Remaining $= 262.5$ cm^2
5	**(a)** 896 cm²	Circle $= 2464$ cm^2; square $= 1568$ cm^2; Difference $= 896$ cm^2
6	**(b)** 850.5 cm²	Circle area $= \frac{22}{7}\times31.5^2 = 3118.5$ cm^2; square $= 3969$ cm^2; Left over $= 850.5$ cm^2
7	**(d)** 1914 m²	Area $= \pi(R^2 - r^2) = \frac{22}{7}(47^2 - 40^2)$; Area of path $= 1914$ m^2
8	**(d)** 38.5 cm²	$\frac{30}{360}\times\frac{22}{7}\times(14^2 - 7^2)$; Area $= 38.5$ cm^2
9	**(c)** 135°	$\theta = \frac{33\times360}{2\times\frac{22}{7}\times14}$; $\theta = 135^\circ$
10	**(b)** 1400 cm²	Sector $= 3850$ cm^2, triangle $= \frac{1}{2}\times70\times70 = 2450$ cm^2; Segment $= 3850 - 2450 = 1400$ cm^2

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Circumference $= 2\times\frac{22}{7}\times7 = 44$ cm; Distance 22 m $= 2200$ cm; revolutions $= \frac{2200}{44}$ (1); Revolutions $= 50$ (1).	1+1
12	In 9 minutes the hand turns through $6^\circ\times9 = 54^\circ$; Area $= \frac{54}{360}\times\frac{22}{7}\times28^2$ (1); Area $= 369.6$ cm^2 (1).	1+1
13	The horse can graze a sector of angle 180° and radius 7 m; Area $= \frac{180}{360}\times\frac{22}{7}\times7^2$ (1); Area $= 77$ m^2 (1).	1+1
14	Circumference $= 2\times\frac{22}{7}\times28 = 176$ cm; Distance 88 m $= 8800$ cm; revolutions $= \frac{8800}{176}$ (1); Revolutions $= 50$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Circumference $= 2\pi r = 2\times\frac{22}{7}\times42 = 264$ m (1); Cost $= 264\times7$ (1); Cost $=$ ₹ 1848 (1).	1+1+1
16	Let the new radius be R; $2\pi R = 2\pi r_1 + 2\pi r_2$ (1); $R = r_1 + r_2 = 21 + 16$ (1); $R = 37$ cm (1).	1+1+1
17	Let the new radius be R; $\pi R^2 = \pi r_1^2 + \pi r_2^2$ (1); $R^2 = 10^2 + 24^2 = 676$ (1); $R = 26$ cm (1).	1+1+1
	OR: Circumference $= 2\pi r = 2\times\frac{22}{7}\times49 = 308$ m (1); Cost $= 308\times5$ (1); Cost $=$ ₹ 1540 (1).	1+1+1
18	Let the new radius be R; $2\pi R = 2\pi r_1 + 2\pi r_2$ (1); $R = r_1 + r_2 = 9 + 4$ (1); $R = 13$ cm (1).	1+1+1
	OR: Let the new radius be R; $\pi R^2 = \pi r_1^2 + \pi r_2^2$ (1); $R^2 = 18^2 + 24^2 = 900$ (1); $R = 30$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Required area $= \frac{\theta}{360}\times\pi(R^2 - r^2)$ (2); $\frac{36}{360}\times\frac{22}{7}\times(14^2 - 7^2)$ (2); Area $= 46.2$ cm^2 (1).	2+2+1
	OR: Grazed area is a quadrant: $\frac{1}{4}\pi r^2$ (2); Increase $= \frac{1}{4}\times\frac{22}{7}(35^2 - 14^2)$ (2); Increase $= 808.5$ m^2 (1).	2+2+1
20	Required area $= \frac{\theta}{360}\times\pi(R^2 - r^2)$ (2); $\frac{24}{360}\times\frac{22}{7}\times(14^2 - 7^2)$ (2); Area $= 30.8$ cm^2 (1).	2+2+1
	OR: Grazed area is a quadrant: $\frac{1}{4}\pi r^2$ (2); Increase $= \frac{1}{4}\times\frac{22}{7}(42^2 - 21^2)$ (2); Increase $= 1039.5$ m^2 (1).	2+2+1

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